University of Engineering and Technology Lahore

Section Course Outline Report

Department: Computer Science Printed Date: April 29, 2026

Section Course Detail	
Semester	Fall 2025
Department	Computer Science
Section	A,B
Subject Title	CS-300 Operating Systems
Subject Domain	Non-Engineering
Subject Knowledge	Humanities
Contact	maslam@uet.edu.pk

Measureable Student Learning Outcomes					
CLOs	Description	PLOs	Domain	Domain Level	Assessments
CLO1	Understand the concept of virtualization that operating systems creates as resource manager to manage physical	PLO02	Cognitive	2. Understand	Quiz 1,Mid,Quiz 2 ,Assignment
CLO2	Analyze the problems of concurrency in multi-threaded environment and propose solutions to avoid deadlocks and	PLO02	Cognitive	4. Analyze	Mid,Final
CLO3	Differentiate between available persistence storages and file systems in terms of usage and performance.	PLO04	Cognitive	5. Evaluate	Mid,Final,Final

Class Timings

Grading Policy
Quiz 1 10.0
Mid 30.0
Quiz 2 10.0
Assignment 10.0
Final 40.0

Section Content

Week (Lec)	Topics	CLO's
week1	Overview, Introduction to basic concepts of Operating Systems, Definitions, Kernel and operating systems basic Architecture	CLO1

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Week (Lec)	Topics	CLO's
week2	Introduction to Operating Systems Structures, User and Operating Systems Interface, HCI in context of Operating Systems and historical development in Operating Systems from inception to the latest Operating Systems.	CLO1
week3	Process: definitions and basic concepts, operations on processes, process states	CLO1
week4	Process synchronization: Principles of concurrency, critical section problem, Monitors, Semaphores, synchronization in various operating system	CLO2, CLO1
week5	Analysis and simulation of concurrent processes on simulator	CLO2
week6	Threads: overview and basic concepts, standard thread libraries	CLO2
week7	Deadlock: Introduction, necessary condition for deadlock, identification, prevention, avoidance	CLO2
week8	Deadlock scenarios with coded examples run on simulator, Deadlock recovery	CLO2
week9	Memory Management: Swapping, Contiguous Memory Allocation, Paging, Segmentation, Page Table	CLO1
week10	Virtual memory Introduction and management techniques	CLO2
week11	File system implementation: structure, log-structured, NFS, free space managemen	CLO3

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Week (Lec)	Topics	CLO's
week12	Mass-Storage Structure: Disk structure, attachment, scheduling, management, RAID structure, Disk attachment, sharing, scheduling.	CLO3
week13	I/O systems, I/O HW, STREAMS, Application I/O subsystems	CLO3
week14	Systems Protection: Access matrix and its implementation, domain of protection, Access rights, protection models.	CLO2
week15	Systems Security: problems and threats, cryptography, user authentication, Firewall	CLO2